

Exercise Sheet 0: Gaussian random variables

In all the exercises, we consider $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space large enough to support all the random variables introduced.

Exercise 1. The density f of a standard Gaussian random variable $X \sim \mathcal{N}(0, 1)$ is given by

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad \forall x \in \mathbb{R}.$$

Let us write

$$I = \left(\int_{\mathbb{R}} e^{-\frac{1}{2}x^2} dx \right)^2.$$

1. Compute I and show that f defines indeed a probability density function.
2. Let $\mu \in \mathbb{R}$ and $\sigma > 0$, compute the density of the random variable $Y := \mu + \sigma X$.
3. We define the moment-generating function M_X of X by

$$M_X(t) := \mathbb{E} [e^{tX}] \quad \forall t \in \mathbb{R}.$$

Give an explicit formula for M_X . Deduce an explicit formula for the moment-generating function M_Y of Y .

Exercise 2. Let X be a random variable with distribution $\mathcal{N}(0, 1)$.

1. Prove by induction that $\mathbb{E} [X^{2k+1}] = 0$ for all $k \in \mathbb{N}$.
2. Prove by induction that $\mathbb{E} [X^{2k}] = \frac{(2k)!}{2^k k!}$ for all $k \in \mathbb{N}$.
3. Show that $\mathbb{E} [X^k] = M_X^{(k)}(0)$ for all $k \in \mathbb{N}$.

Exercise 3. Let $(X_n)_n$ be a sequence of centered i.i.d. random variables with variance $0 < \sigma^2 < +\infty$. We define the sequence $(Z_n)_n$ by

$$Z_n = \frac{1}{\sigma\sqrt{n}} \sum_{k=1}^n X_k.$$

By the central limit theorem, we know that

$$\lim_{n \rightarrow +\infty} \mathbb{E} [e^{itZ_n}] = e^{-\frac{1}{2}t^2} \quad \forall t \in \mathbb{R}.$$

1. Let $D_n = Z_{2n} - Z_n$ for all $n \in \mathbb{N}$. Compute the characteristic function of D_n and show that it converges in distribution.
2. By considering $\mathbb{P}(|D_n| \geq \varepsilon)$, show that Z_n does not converge in probability.

Exercise 4. Let $(X, Y)^\top$ be a centered Gaussian vector with covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}.$$

1. Compute the characteristic function of $(X, Y)^\top$.
2. Show that if $\rho = 0$, then X and Y are independent.
3. Is it still true in general if X and Y are not Gaussian? Give an example.

Exercise 5. Let $(X, Y)^\top$ be a centered Gaussian vector with covariance matrix

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

1. Compute $\mathbb{E}[X|Y]$.
2. Compute $\mathbb{V}(X|Y)$.
3. Deduce the conditional distribution of $X|Y$.

Exercise 6. Let X be a random variable with distribution $\mathcal{N}(0, 1)$. Give the probability density function of the random variable $Y := e^X$.

Exercise 7. Let X be a random variable with distribution $\mathcal{N}(\mu, \sigma^2)$, and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a bounded C^1 function with bounded derivative. Show that

$$\mathbb{E}[(X - \mu)g(X)] = \sigma^2 \mathbb{E}[g'(X)].$$

Exercise 8. Let $(\xi_n)_n$ be a sequence of i.i.d. random variables with distribution $\mathcal{N}(0, 1)$. We define the sequence $(S_n)_n$ by

$$S_n = \sum_{k=1}^n \xi_k.$$

1. Compute the distribution of S_n .
2. Compute $\text{Cov}(S_n, S_m)$ for all $n, m \in \mathbb{N}^*$.
3. Give the distribution of $S_n - S_m$ for $1 \leq m \leq n$, and show that it is independent from $\sigma(\xi_1, \dots, \xi_m)$.